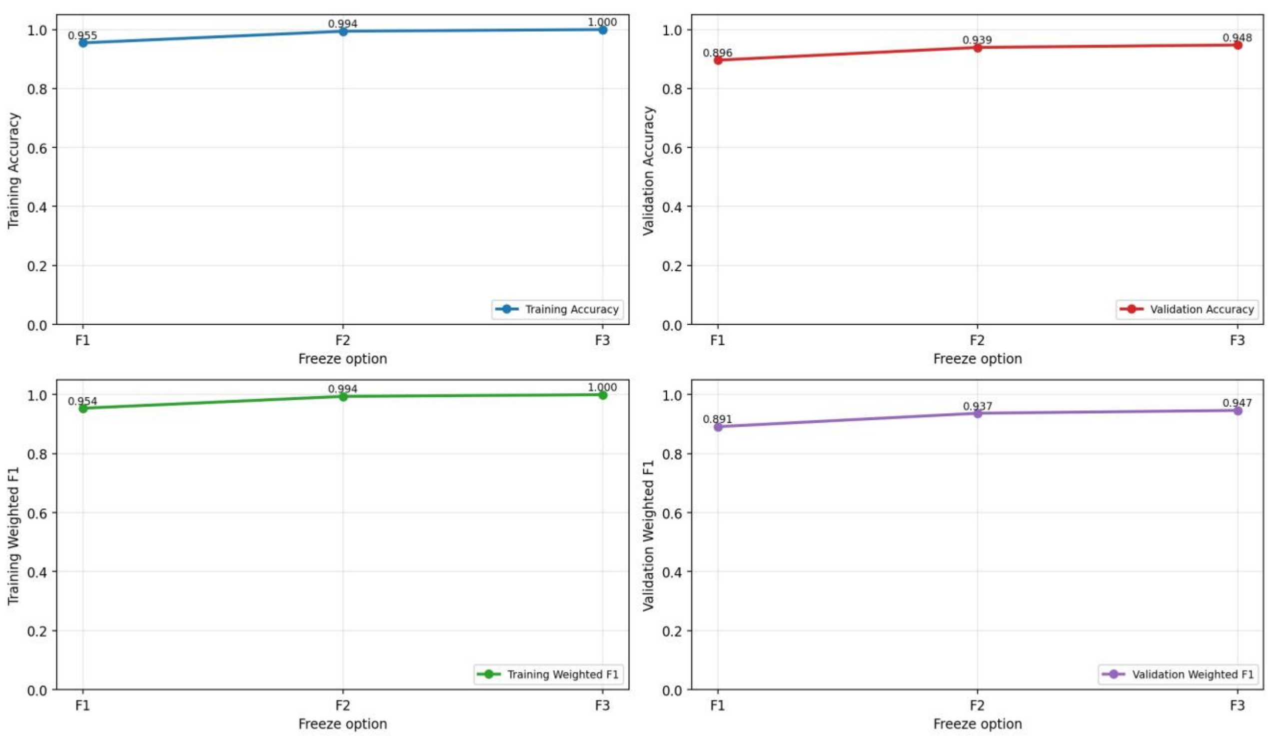
Objective, Baselines, and RGB Transfer Learning

This report completes the final project requirements after Report 1. The task is eight-class object recognition for autonomous driving
using RGB and infrared patches. Report 1 established the custom CNN baselines; this report adds RGB VGG16 transfer learning, IR autoencoder features, sensor comparison, misclassification analysis, and final model selection.

Model	Train Acc	Val Acc	Train F1	Val F1
RGB Part 1 Arch D	0.9999	0.9201	0.9999	0.9185
IR Part 1 Arch D	0.9766	0.8880	0.9772	0.8792

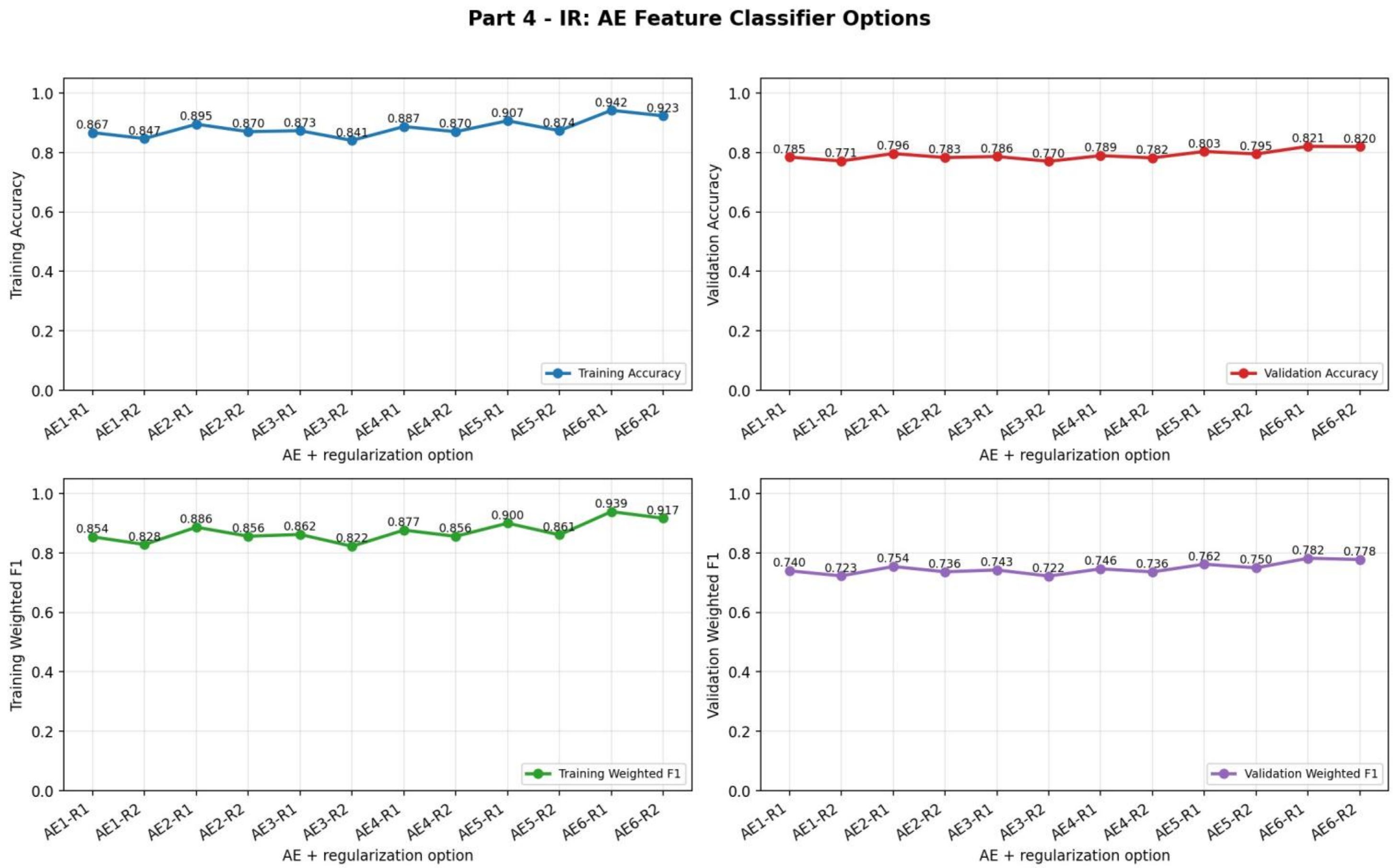
Freeze	Train Acc	Val Acc	Train F1	Val F1
F1	0.9547	0.8961	0.9539	0.8914
F2	0.9941	0.9392	0.9941	0.9370
F3	0.9999	0.9475	0.9999	0.9466

Part 3 - RGB: VGG16 Freeze-Level Comparison



IR Autoencoder Feature Classifiers

The IR autoencoder experiment used six encoder configurations and two dense-head regularization settings, for 12 total options. The best AE option was AE6-R1 with validation F1 0.7820. The supervised IR Part 1 baseline validation F1 was 0.8792.



Final Model Selection and Refinement

Candidate	Train Acc	Val Acc	Train F1	Val F1
RGB final	0.9955	0.9498	0.9955	0.9480
Best RGB VGG16	0.9999	0.9475	0.9999	0.9466
RGB Part 1 Arch D	0.9999	0.9201	0.9999	0.9185
IR final	0.0000	0.9263	0.0000	0.9227
Best IR AE	0.9420	0.8206	0.9389	0.7820
IR Part 1 Arch D	0.9766	0.8880	0.9772	0.8792

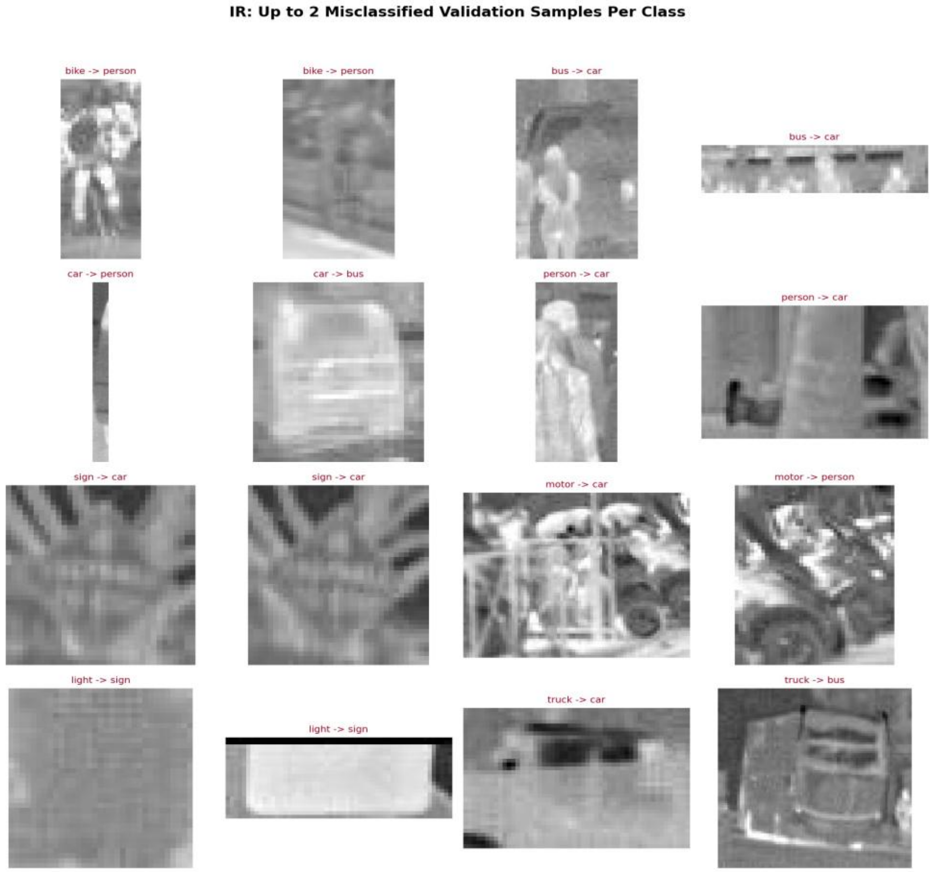
- Final RGB model: RGB VGG16 refined (validation F1 0.9480).
- Final IR model: IR CNN Arch D + Aug L2 (validation F1 0.9227).
- Blind-test strategy: select by validation weighted F1 and avoid using withheld labels. Transfer learning and AE results are retained even when the custom CNN remains stronger.
- IR AE did not come within 0.02 F1 of IR Part 1 Arch D; final IR uses the stronger supervised CNN reference.

Observed reliability pattern: RGB generally benefits from color and texture detail for traffic signs, lights, and object boundaries. Infrared can be more stable where visible color/texture is degraded, but small or low-resolution classes remain difficult in both sensors. The paired examples on page 5 show cases where the sensors disagree.

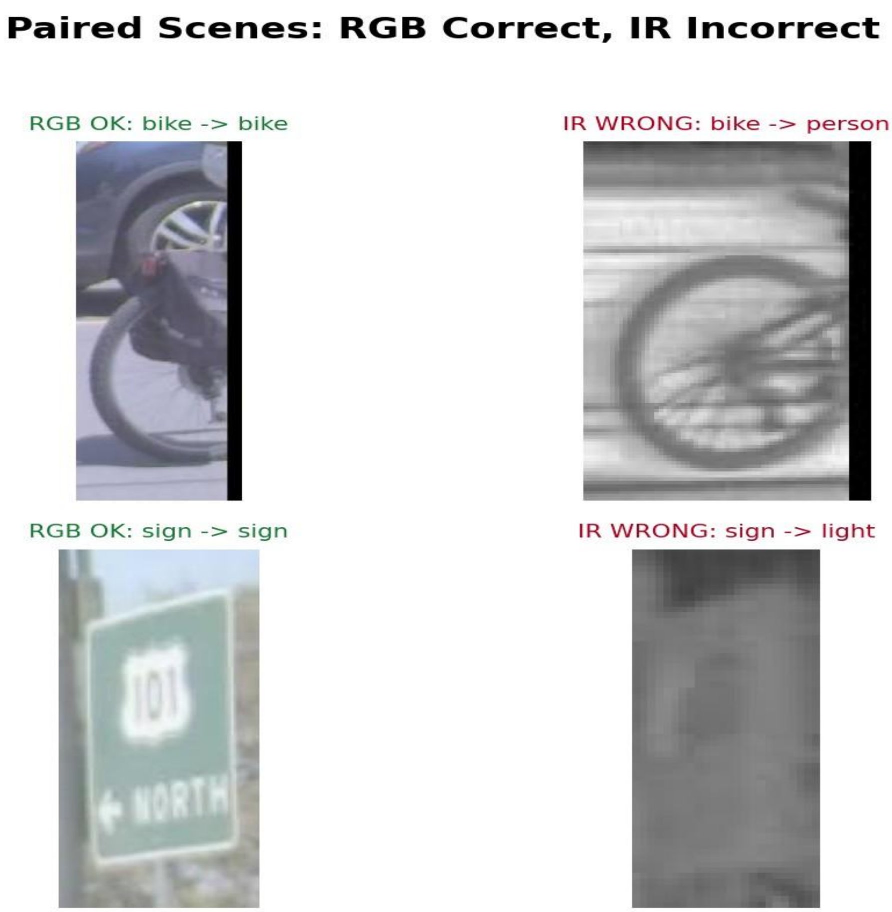
Misclassified Samples By Sensor

The grids below show up to two validation misclassifications per class where available. Counts are tracked in the generated Part 5 JSON summary so classes with fewer than two misses are explicitly documented.

RGB: Up to 2 Misclassified Validation Samples Per Class



RGB-vs-IR Sensor Disagreement Analysis



Paired examples found: RGB-correct/IR-wrong = 2, IR-correct/RGB-wrong = 3. When exact same-class paired examples were unavailable, the notebook used same-frame paired-scene examples and labels them as scene-level comparisons.